Indian Institute of Technology Ropar

Department of Mechanical Engineering
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CP302 - Capstone Project I

End-Semester Report

Simulation, Control, and State Estimation for Cooperative Transportation of a Rod-Shaped Payload Using Two Quadrotors

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1 Problem Definition

Unmanned aerial vehicles (UAVs), particularly quadrotors, have emerged as versatile platforms for a wide range of applications including logistics, disaster response, infrastructure inspection, and construction. Their ability to perform vertical take-off and landing, hover, and maneuver in confined environments makes them suitable for transporting objects in areas that may be inaccessible to ground vehicles. However, the payload capacity of a single quadrotor is inherently limited due to constraints in thrust generation, battery capacity, and flight stability.

Many practical transportation tasks involve objects that are large, heavy, or geometrically extended, making them unsuitable for transport by a single aerial vehicle. In such scenarios, cooperative transportation using multiple UAVs becomes a viable solution. By distributing the load among multiple vehicles, the system can increase payload capacity while maintaining maneuverability and operational flexibility.

This project focuses on the cooperative transportation of a rigid rod-shaped payload using two quadrotors. The primary objective is to develop and analyze a control framework that enables the two UAVs to transport the payload while maintaining stable flight, coordinated motion, and controlled payload orientation. The problem involves challenges related to nonlinear system dynamics, coupling between the UAVs and the payload, and the design of cooperative control strategies [1].

1.1 Control of single Quadrotor UAVs

Quadrotors are underactuated aerial systems with six degrees of freedom (three translational and three rotational) but only four independent control inputs generated by the propellers. The control inputs typically consist of total thrust and three rotational torques corresponding to roll, pitch, and yaw motions. The nonlinear dynamics of a quadrotor can be generally represented as

$$m\ddot{\mathbf{p}} = R(\eta)\mathbf{f}_T - m\mathbf{g} \tag{1}$$

$$J\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times J\boldsymbol{\omega} = \boldsymbol{\tau} \tag{2}$$

where $\mathbf{p}$ represents the position of the UAV in the inertial frame, $R(\eta)$ is the rotation matrix determined by the attitude angles, $\mathbf{f}_T$ is the thrust force vector, J is the inertia matrix, and $\boldsymbol{\tau}$ denotes the control torques. Due to their nonlinear and coupled dynamics, quadrotors are sensitive to disturbances and modeling uncertainties.

For payload transportation tasks, the controller must ensure stable flight while maintaining precise trajectory tracking and attitude stabilization. Additionally, it must compensate for external disturbances introduced by the payload dynamics.

1.2 Payload Transportation Dynamics

When a payload is attached to a UAV, the dynamics of the overall system become coupled. The payload introduces additional forces, moments, and inertial effects that alter the motion of the aerial vehicle. Depending on the attachment mechanism, such as rigid mounting, cable suspension, or grasping mechanisms, the resulting system dynamics can vary significantly.

In this project, the payload is modeled as a rigid rod. Rod-shaped payloads are commonly encountered in real-world scenarios such as transporting pipes, structural beams, or elongated mechanical components. The elongated geometry of the rod introduces additional rotational dynamics and increases the complexity of maintaining stable orientation during transportation. Any imbalance or disturbance can cause oscillations, which may destabilize the UAVs if not properly controlled.

1.3 Payload State estimation

Accurate payload state estimation is essential for stable cooperative transportation using multiple UAVs. During flight, the payload may experience oscillations and orientation changes due to disturbances, acceleration of the UAVs, and coupling effects between the vehicles and the payload. Without proper estimation of payload position, velocity, and swing angles, the controller may fail to maintain stable transportation and coordinated motion.

Traditional payload estimation approaches often rely on external sensing systems such as motion capture cameras or payload-mounted sensors [2]. Although these methods provide accurate measurements, they increase system complexity, cost, and payload weight. For practical outdoor applications, onboard estimation methods using IMUs, force sensors, and UAV state measurements are more suitable.

Accurate payload state estimation improves trajectory tracking, suppresses oscillations, and enhances the overall stability of the cooperative transportation system.

1.4 Cooperative Transportation of Rod-Shaped Payload

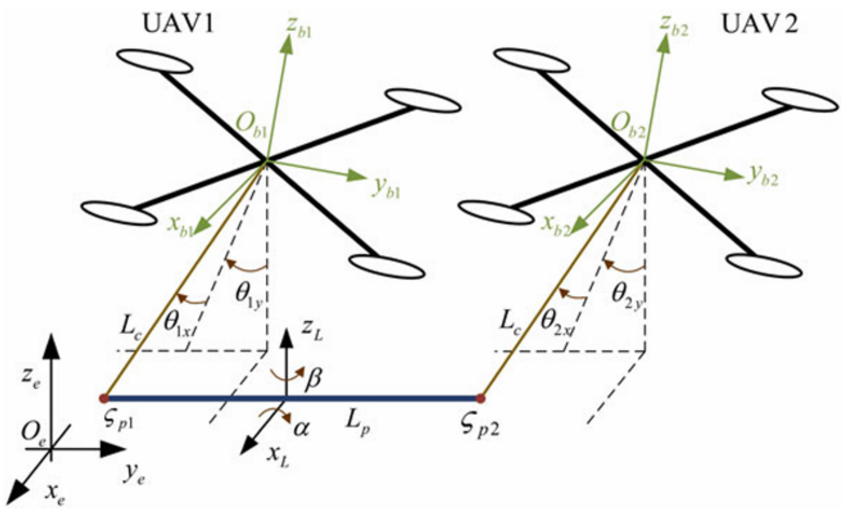

Figure 1: System configuration for cooperative transportation of a rod-shaped payload using two quadrotors. Each UAV is connected to the payload through a cable of length L_c. The payload of length L_p is attached at points ζ_{p1} and ζ_{p2}. The inertial frame (x_e, y_e, z_e) and body frames of each UAV (x_{b1}, y_{b1}, z_{b1}) and (x_{b2}, y_{b2}, z_{b2}) are shown along with the swing angles $\theta_{1x}, \theta_{1y}, \theta_{2x}, \theta_{2y}$ and payload orientation angles α and β. [3]

The cooperative transportation of a rod-shaped payload using two quadrotors introduces several additional challenges compared to single-UAV payload transport. The payload must remain stable while both UAVs generate coordinated forces and moments to maintain the desired trajectory. The motion of each UAV influences the dynamics of the payload as well as the other UAV, resulting in a strongly coupled multi-agent system.

The control strategy must address several important aspects:

- Coordinated motion between the two UAVs
- Load sharing and balanced force generation
- Stability of the payload orientation
- Suppression of oscillations during flight

The objective of this project is therefore to study and implement a control framework that enables two quadrotors to cooperatively transport a rigid rod-shaped payload while maintaining system stability and trajectory tracking performance. The proposed framework will involve modeling the coupled UAV–payload dynamics and validating the control approach through simulation.

2 Literature Review

2.1 Control of Quadcopter

2.1.1 Overview of Quadcopter Control Strategies

Quadrotor unmanned aerial vehicles (UAVs) have attracted significant research interest due to their mechanical simplicity, vertical takeoff and landing capability, and maneuverability. However, quadrotor systems are inherently nonlinear, underactuated, and highly coupled, making control design a challenging task.

A comprehensive overview of quadrotor control techniques is presented by Asignacion and Satoshi [4], which surveys the historical development and current landscape of autonomous quadrotor control. Early control approaches primarily relied on cascaded PID controllers, where an inner-loop attitude controller and an outer-loop position controller are used to stabilize the vehicle. These controllers remain widely used due to their simplicity and ease of implementation.

More advanced control strategies have been developed to address the nonlinear dynamics and external disturbances affecting quadrotor systems. These include feedback linearization, backstepping control, sliding mode control, disturbance observer-based control, and adaptive control methods [4]. Among these, sliding mode and backstepping controllers have demonstrated strong robustness and tracking performance in the presence of modeling uncertainties.

In recent years, optimal and learning-based control approaches have also gained attention. Techniques such as Model Predictive Control (MPC), Linear Quadratic Regulation (LQR), and reinforcement learning-based controllers have been explored to improve trajectory tracking accuracy and adaptability to changing environments. Despite these advances, classical controllers such as PID remain popular in practical implementations due to their computational efficiency and reliability.

2.2 Aerial Vehicles with Variable Structures

A detailed presentation on these reconfigurable aerial vehicle architectures was prepared and discussed with the project supervisors during the early phase of this work. The presentation slides summarizing the explored designs and their control challenges can be accessed at:

Reconfigurable Aerial Vehicles Literature Exploration

2.2.1 Overview of Reconfigurable and Morphing Quadrotors

In addition to improvements in control strategies, recent research has explored aerial vehicles with variable or reconfigurable structures to enhance maneuverability and task capability. Morphing aerial robots are capable of changing their geometry during flight, allowing them to adapt to different operational requirements such as navigating confined spaces, grasping objects, or improving aerodynamic performance.

Several works have proposed innovative reconfigurable quadrotor designs. Examples include the Ring-Rotor system capable of aerial grasping and transportation, quadrotors with deformable frames allowing changes in geometry without thrust loss, midair-reconfigurable quadcopters using passive hinges, and foldable drones capable of compressing their structure to pass through narrow openings. These designs demonstrate how structural reconfiguration can expand the functional capabilities of aerial robots beyond traditional rigid-frame quadrotors.

Although these morphing platforms offer promising capabilities, they introduce additional complexities in modeling and control due to changes in inertia, thrust directions, and vehicle geometry during flight. Consequently, many of these systems require adaptive or model-based control frameworks capable of handling time-varying dynamics.

2.3 Rod-Shaped Payload Transportation

2.3.1 Cooperative Control for Rod-Shaped Payload Transportation with Two Quadrotors

Wang and Chen [3] investigate cooperative transportation of a rigid rod-shaped payload using two quadrotors connected through taut cables. The authors formulate the system dynamics using the Euler–Lagrange framework considering both UAV positions and payload swing angles as generalized coordinates. The system is modeled with nine degrees of freedom defined by

$$q = \begin{bmatrix} x_1 & y_1 & z_1 & \theta_{1x} & \theta_{1y} & \alpha & \beta & \theta_{2x} & \theta_{2y} \end{bmatrix}^T \tag{3}$$

where (x_i, y_i, z_i) represent UAV positions, θ_{ix}, θ_{iy} denote cable swing angles, and (α, β) represent the payload orientation. The dynamics are derived using the Lagrangian formulation

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) - \frac{\partial L}{\partial q} = Q \tag{4}$$

where $L = T - V$ is the Lagrangian, with T and V representing kinetic and potential energies respectively. This results in the nonlinear dynamic model

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = Q \tag{5}$$

where $M(q)$ is the inertia matrix, $C(q, \dot{q})$ represents Coriolis and centrifugal effects, and $G(q)$ denotes gravitational forces.

To regulate the system and suppress payload oscillations, the authors propose a nonlinear model predictive control (NMPC) scheme. The controller predicts the future system states over a finite horizon and minimizes a quadratic cost function of the form

$$J = \sum_{i=0}^{N_p-1} e_{x,i}^T Q e_{x,i} + e_{x,N}^T Q_N e_{x,N} \tag{6}$$

where $e_{x,i}$ represents the state tracking error and Q, Q_N are weighting matrices. The optimization is solved at each control step subject to thrust constraints and system dynamics. This predictive formulation allows simultaneous regulation of quadrotor positions and payload swing angles. Simulation results demonstrate that the NMPC controller achieves faster convergence and significantly smaller payload oscillations compared with conventional PD controllers.

2.3.2 Rod-Shaped Payload Transportation Using Multiple Quadrotors

Villa et al. [5] address the cooperative transportation of a rod-shaped payload by modeling the UAV–payload system as a virtual structure formation. In this approach, the payload and the two quadrotors are treated as a single rigid formation whose configuration is described by the vector

$$q = \begin{bmatrix} x_f & y_f & z_f & \alpha & \beta & \rho \end{bmatrix}^T \tag{7}$$

where (x_f, y_f, z_f) represent the position of one endpoint of the formation, α and β define the orientation of the rod, and ρ denotes its length.

The controller first regulates the formation variables using a nonlinear saturation-based control law

$$\dot{q}_r = \dot{q}_d + K_1 \tanh(K_2 \tilde{q}) \tag{8}$$

where $\tilde{q} = q_d - q$ represents the formation error and K_1, K_2 are positive gain matrices. The hyperbolic tangent term limits the control signals and improves robustness against actuator saturation.

Once the formation reference is computed, the desired positions of the quadrotors are obtained from the load pose through inverse geometric transformations. Each UAV then tracks these references using an inner-loop controller derived from partial feedback linearization of the quadrotor dynamics

$$F = \bar{M}_{aa}\eta_a + \bar{E}_a \tag{9}$$

where η_a contains tracking error terms and control gains. The resulting closed-loop dynamics ensure that the tracking errors converge to zero asymptotically. Experimental validation using two AR.Drone platforms demonstrates stable transportation of an aluminum rod with acceptable tracking accuracy and disturbance rejection.

2.3.3 A Control Strategy for Swing Suppression in Dual-UAV Cooperative Rod Payload Transportation

Xu et al. [6] propose a hierarchical control framework for suppressing oscillations in a dual-UAV suspended payload transportation system. The dynamic model of the system is derived using the Euler–Lagrange method, considering UAV positions, attitudes, and cable swing angles as generalized coordinates. The system dynamics can be expressed in compact form as

$$M(q)\ddot{q} + V(q, \dot{q})\dot{q} + G(q) = F \tag{10}$$

where $M(q)$ represents the inertia matrix, $V(q, \dot{q})$ the Coriolis matrix, and $G(q)$ the gravitational forces acting on the system.

To address payload oscillations, the authors design a hierarchical controller consisting of a position loop and an attitude loop. In the outer loop, an Integral Backstepping Adaptive Anti-Swing Controller (IBAASC) is introduced to regulate UAV position while compensating for payload swing dynamics. The tracking error for the UAV position is defined as

$$e_x = x_d - x \tag{11}$$

and a Lyapunov-based control law is derived to ensure asymptotic convergence of both position errors and swing angles. An adaptive disturbance estimator is also incorporated to compensate for unknown external disturbances.

For attitude stabilization, a Nonsingular Terminal Sliding Mode Controller (NTSMC) is employed. The sliding surface is defined as

$$s = \dot{e} + \beta e^{q/p} \tag{12}$$

where e denotes the attitude tracking error and p, q are odd integers satisfying $1 < q/p < 2$. The controller guarantees finite-time convergence of attitude errors while avoiding singularities commonly encountered in terminal sliding mode control. Simulation results demonstrate that the proposed control strategy significantly reduces payload swing angles and improves trajectory tracking accuracy compared with conventional PID control approaches.

2.4 Reinforcement Learning for Payload Stabilization

Cable-suspended payload transportation introduces nonlinear and underactuated dynamics that make payload stabilization challenging. Conventional model-based controllers often require accurate system models and may perform poorly under uncertainties and disturbances. Reinforcement Learning (RL) has emerged as an effective alternative by learning control policies directly through interaction with the environment. RL-based approaches can reduce payload oscillations while improving flight stability and robustness.

2.4.1 A Study on Reinforcement Learning based Control of Quadcopter with a Cable-suspended Payload

Prajapati et al. [7] investigate the use of Reinforcement Learning for stabilizing a quadcopter carrying a cable-suspended payload. The coupled quadrotor–payload dynamics are modeled as

$$(m_q + m_p)(\dot{v}_q + ge_3) = [qfRe_3 - m_q l(\dot{q} \cdot \dot{q})]q + d_p \tag{13}$$

where m_q and m_p are the quadrotor and payload masses, f is thrust, and R is the rotation matrix. The rotational dynamics are represented by

$$J\dot{\Omega} + \Omega \times J\Omega = M + d_\tau \tag{14}$$

The authors employ Proximal Policy Optimization (PPO) to learn a control policy that minimizes payload swing and tracking error. The reward function is defined as

$$r_t = -\left(w_1 e_p^2 + w_2 e_v^2 + w_3 e_q^2 + w_4 u^2\right) \tag{15}$$

where the terms represent position, velocity, swing, and control effort errors. Simulation results demonstrate that the PPO-based controller effectively reduces payload oscillations and improves flight stability.

2.5 Payload state estimation/Swing angle estimation

Accurate estimation of payload motion and swing angle is essential for stable aerial transportation. Since external sensing systems increase complexity and cost, recent approaches focus on estimating payload states using onboard UAV sensors such as IMUs and force sensors. Kalman filtering and observer-based methods are commonly used to reconstruct payload motion under nonlinear and uncertain dynamics.

2.5.1 Payload State Estimation for Cooperative Manipulation Using Multiple UAVs

Ohira et al. [8] propose a payload state estimation framework using an Unscented Kalman Filter (UKF) for cooperative aerial manipulation. The tether direction is represented using spherical coordinates as

$$u_r = \begin{bmatrix} \sin\theta_s \cos\phi_{abs} \\ \sin\theta_s \sin\phi_{abs} \\ -\cos\theta_s \end{bmatrix} \tag{16}$$

The tether angular dynamics are expressed as

$$\ddot{\theta}_s = -\frac{g}{l}\sin\theta_s + \frac{R_f F_f \cdot u_\theta}{ml} \tag{17}$$

Using onboard accelerometer measurements, the tether force is estimated through

$$F_s = R_f(F_f - ma_b) \tag{18}$$

The UKF-based estimator accurately reconstructs tether angles and payload motion without payload-mounted sensors.

2.5.2 Swing angle estimation for multicopter slung load applications

De Angelis [9] present a swing angle estimation approach for multicopter slung-load systems using onboard IMU measurements. The payload position is modeled as

$$p_l = p + L \begin{bmatrix} \sin\zeta \\ -\sin\xi\cos\zeta \\ \cos\xi\cos\zeta \end{bmatrix} \tag{19}$$

The coupled system dynamics are represented using the Euler–Lagrange formulation

$$M(\lambda)\dot{\mu} + C(\lambda,\mu)\mu + G(\lambda) = \tau \tag{20}$$

To estimate swing angles, a Fading Gaussian Deterministic (FGD) filter is employed:

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k(y_k - H\hat{x}_{k|k-1}) \tag{21}$$

The proposed estimator improves robustness against modeling uncertainties and accurately reconstructs payload oscillations using only onboard sensing.

3 Progress

During the initial phase of the project, time was primarily spent understanding the dynamics and control challenges associated with aerial payload transportation systems. Literature related to quadrotor control, suspended payload dynamics, nonlinear control, cooperative UAV systems, and reinforcement learning-based aerial control was studied to identify suitable approaches for the project. Based on this study, the problem statement was refined toward *dual-UAV cooperative transportation of a rigid rod-shaped payload*.

A significant portion of the early work was also devoted to building the theoretical background required for advanced control methods. Topics such as state-space modeling, optimal control, LQR, MPC, nonlinear system dynamics, and payload-coupled aerial systems were studied through online lectures, research papers, and control system references. This phase helped establish the mathematical and control foundation necessary for developing cooperative transport controllers.

3.1 Simulation Progress

3.1.1 Initial Simulation Attempts

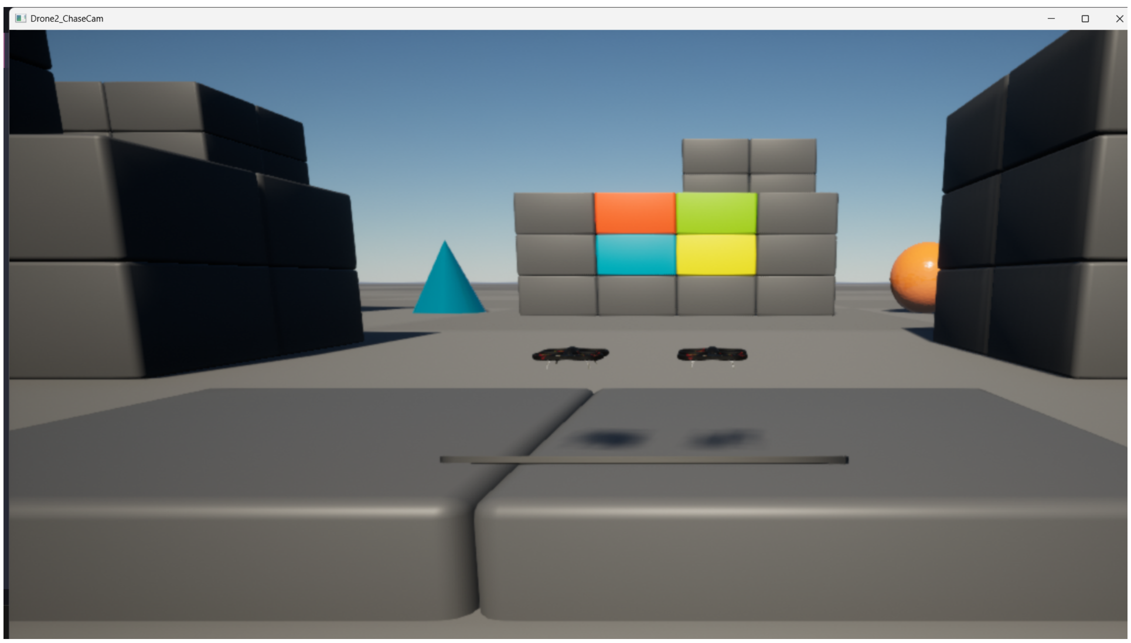

Figure 2: Experiment conducted in AirSim to mimic the pickup of a rod-shaped payload. Two drones are controlled simultaneously while a rod object is placed between them to approximate cooperative transportation behaviour. This setup was used only for visual experimentation and did not realistically simulate physical interaction between the drones and the payload.

The initial simulation exploration was performed using AirSim in order to quickly prototype multi-drone cooperative behavior. Basic dual-drone operation and follower-type coordination were successfully implemented. However, realistic payload transportation using physical constraints between the drones and the payload proved difficult because AirSim does not directly support flexible tether modeling or cooperative rigid-body payload interaction [10].

Several approaches such as artificial payload attachment, collision-based hooks, and simplified rod placement between the drones were explored. Although these approaches enabled visual experimentation, they were not suitable for realistic payload dynamics and cooperative control research. Due to these limitations, the simulation framework was later migrated to a ROS-based PX4–Gazebo environment, which provides significantly better support for robotics simulation, multi-agent coordination, joint modeling, and sensor integration.

3.1.2 PX4-ROS-Gazebo Simulation Environment Setup

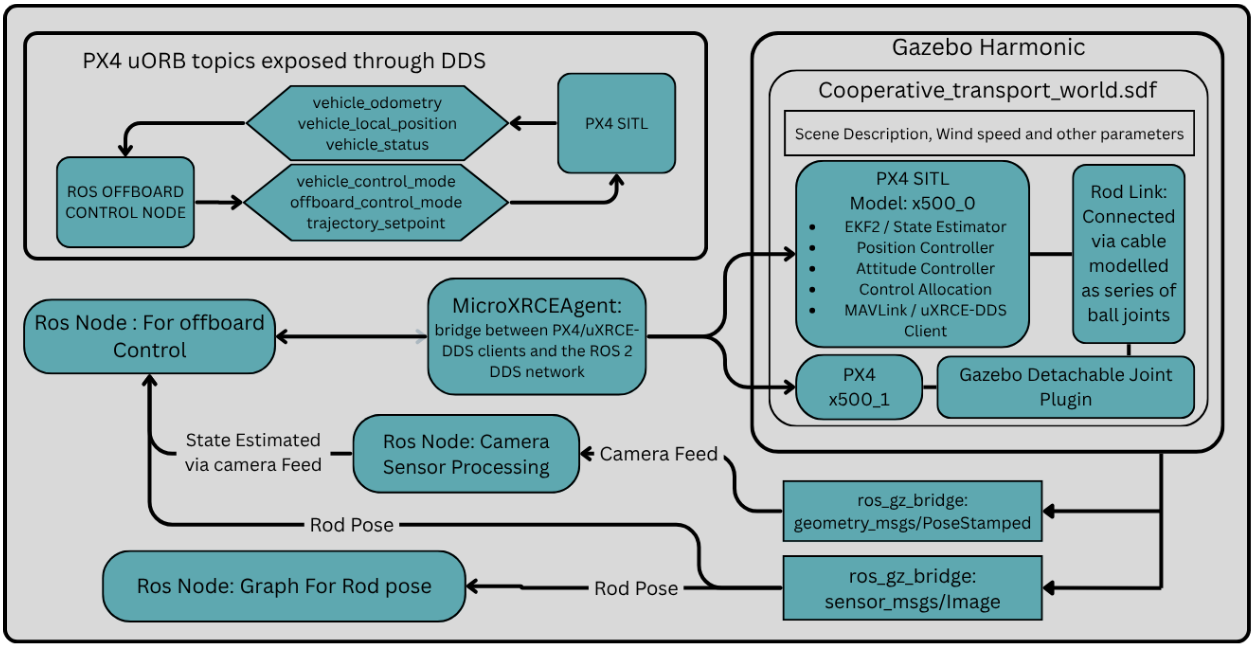

Figure 3: Flow architecture of the simulation framework showing PX4 SITL, Gazebo Harmonic, ROS 2 nodes, MicroXRCE-DDS communication, payload state estimation, and monitoring pipeline for dual-quadrotor cooperative control.

The main simulation framework was developed in **Gazebo Harmonic** with **PX4 SITL** running two cooperative quadrotor vehicles and a rod-shaped payload connected through a simplified cable model. The complete architecture was organized so that the flight stack, simulation world, sensor interfaces, state estimation, and control nodes could interact in real time through ROS 2 and DDS-based communication. Figure 3 summarizes the overall flow of information between PX4, Gazebo, and the custom ROS nodes used for cooperative payload transportation [11–13].

ROS-PX4 communication and offboard control The communication between ROS 2 and PX4 was implemented using `px4_ros_com` together with `microXRCEAgent`. In this setup, PX4 publishes and subscribes to `uORB` topics through the Micro XRCE-DDS client, while the ROS 2 side receives the corresponding DDS messages through the agent bridge. For offboard operation, a dedicated ROS 2 control node was developed to publish the `OffboardControlMode` and `TrajectorySetpoint` messages and to subscribe to feedback topics such as `VehicleOdometry`, `VehicleLocalPosition`, and `VehicleStatus`. The offboard node continuously streams setpoints to PX4 so that the vehicle remains in offboard mode and does not trigger failsafe behavior due to setpoint timeout. This communication layer forms the core interface between the high-level cooperative controller and the low-level PX4 flight controller.

Dual-drone spawning and custom world configuration Both quadrotors were spawned inside a custom `cooperative_transport_world.sdf` file. The two vehicles were created from the same `x500` model but instantiated with different model names, namely `x500_0` and `x500_1`, so that each drone could run as an independent PX4 SITL instance while still sharing the same physical and sensor configuration. The world file was also extended to include the rod payload, scene parameters, wind settings, and the required plugins for cooperative transport. Using a single custom world file made it easier to manage the relative placement of the vehicles, payload geometry, bridge interfaces, and simulation parameters in one place.

Payload and cable modeling The rod and cable were modeled specifically for cooperative transport experiments rather than full structural fidelity. The cable was represented as a chain of rigid segments connected using **ball joints**, which provided an efficient approximation of a flexible tether while keeping the simulation stable and computationally tractable. For a cable of approximately 5 m length, five ball joints were used in the model. This approach was preferred over a more complex continuum or finite-element cable representation because it is easier to simulate in real time and is sufficient for studying the closed-loop transport behavior of the coupled drone-payload system. To prevent the model from behaving as a tree-structured mechanism and to allow controlled separation when needed, the `Gazebo Detachable Joint Plugin` was used.

Initial spawn and collision handling At the beginning of simulation, the drones and the suspended payload had to be positioned carefully to avoid initialization failures. Since the focus of this work is on cooperative control rather than ground-contact dynamics, the collision interactions between the rod, cable, and ground were removed. This allowed the two quadrotors to be spawned on the ground while the rod and cable extended below the ground plane during initialization without producing unstable contact forces or launch errors. Alternative approaches, such as enforcing strict collision constraints during spawn, caused practical difficulties in starting the simulation reliably. Removing unnecessary collision interaction simplified the initialization process and improved robustness of the simulation setup.

Rod pose monitoring and payload state visualization To monitor the payload motion during simulation, the pose of the rod was exposed from Gazebo using `ros_gz_bridge` and mapped to a ROS 2 `geometry_msgs/PoseStamped` topic. A custom ROS node was then developed to track the rod pose, store the incoming state, and visualize the payload trajectory using a graph-based display. This monitoring pipeline was useful for analyzing swing behavior, transport stability, and the effect of control inputs on the payload orientation and position. It also provided a direct feedback path for evaluating the performance of the cooperative controller during experiments.

Camera feed integration and perception pipeline The standard monocular configuration of the `x500` model did not provide the required camera output for our perception pipeline, so a custom camera sensor plugin was added to the drone model. The rendered camera stream was then transferred to ROS 2 using `ros_gz_bridge` and published as `sensor_msgs/Image`. A separate ROS node handled camera processing and state extraction from the visual feed. This node served as the perception interface for estimating the payload state from the onboard camera data and for providing an additional measurement source beyond the simulated odometry and pose topics.

Keyboard-based setpoint generation A keyboard control node was implemented to provide manual reference inputs during testing and debugging. This node accepted user inputs for the desired payload or transport target in terms of x, y, z, yaw, and pitch, and published these commands to the central controller. The main controller subscribed to this setpoint topic and generated the corresponding cooperative transport commands for the two UAVs. This interface was useful for verifying the end-to-end behavior of the system before integrating more advanced autonomy and learning-based control strategies.

Overall system architecture The complete architecture contains three major layers: the **simulation layer**, the **communication layer**, and the **control/perception layer**. In the simulation layer, Gazebo Harmonic hosts the dual-`x500` quadrotor models, the rod payload, the cable mechanism, and the custom plugins inside the `cooperative_transport_world`. In the communication layer, `microXRCEAgent`, `px4_ros_com`, and `ros_gz_bridge` connect PX4, Gazebo, and ROS 2 through DDS topics. In the control/perception layer, custom ROS nodes implement offboard control, camera processing, rod pose tracking, keyboard input handling, and setpoint generation. The architecture allows the cooperative transport system to be tested in a modular way, where each component can be validated independently while still supporting the full closed-loop multi-UAV payload transport pipeline.

Advantages of the developed setup This simulation environment offers several practical advantages. First, it reproduces the essential dynamics of cooperative payload transportation while remaining computationally efficient enough for repeated experiments. Second, it cleanly separates flight control, perception, and payload monitoring into independent ROS 2 nodes, which improves modularity and debugging. Third, the use of PX4 SITL with Gazebo Harmonic enables realistic offboard-control testing under sensor feedback, actuator constraints, and networked communication. Finally, the custom world file and bridge-based architecture make it straightforward to extend the framework later for state estimation, robust control, reinforcement learning, and fault-tolerant cooperative transport experiments.

3.1.3 Implemented PID-Based Cooperative Control for Dual-Quadrotor Payload Transportation

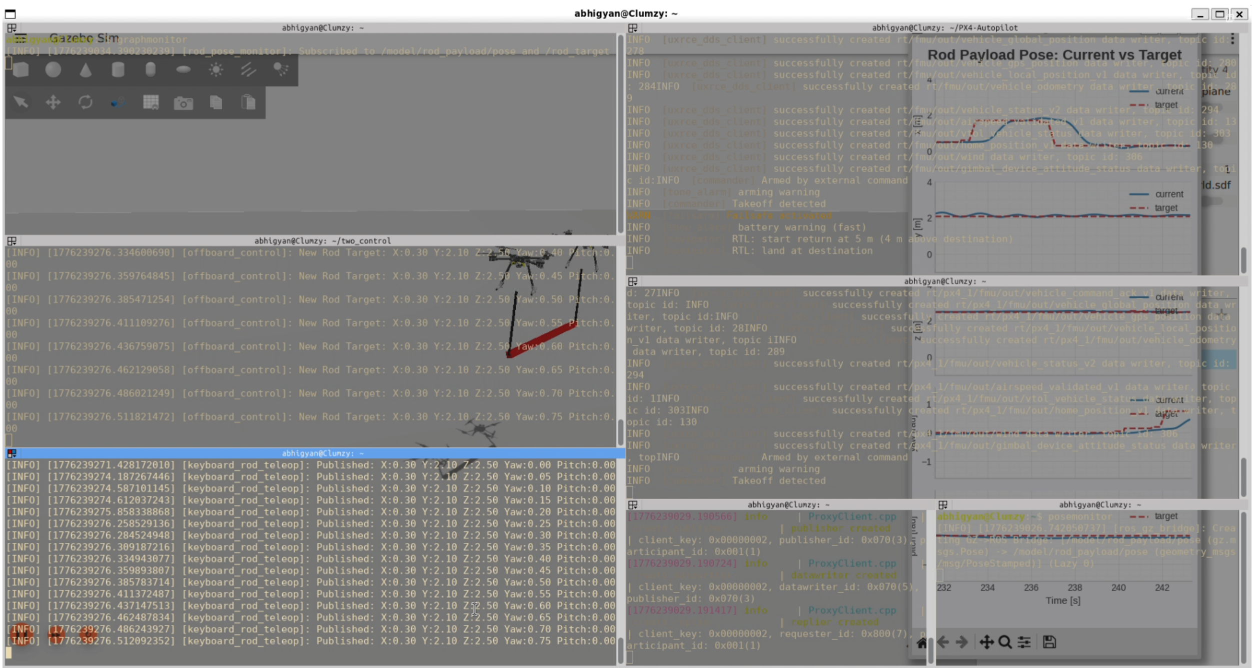

Figure 4: Implementation and real-time monitoring of PID-based cooperative control inputs for dual quadrotor rod-payload transportation in PX4 SITL and Gazebo Harmonic, including payload pose tracking, telemetry visualization, and ROS 2 feedback monitoring.

A closed-loop PID-based cooperative control framework was implemented in ROS 2 for transporting a rod-shaped payload using two quadrotor UAVs operating in PX4 offboard mode. The controller was developed as a custom ROS 2 node in C++ and interfaced with PX4 through `px4_ros_com` and Micro XRCE-DDS communication. The complete controller architecture combines payload pose feedback, cooperative geometric transformation, and decentralized trajectory generation for both UAVs.

Control objective The primary objective of the controller is to regulate the position and orientation of the suspended rod payload rather than directly controlling the individual UAV positions. The desired payload state consists of:

$$[x,\ y,\ z,\ \psi,\ \theta]$$

where x, y, z represent the desired payload position, ψ represents yaw angle, and θ represents the rod pitch angle. The controller computes the corresponding cooperative setpoints for both drones such that the payload center tracks the commanded trajectory while minimizing oscillation and orientation error.

Payload state feedback The actual payload pose was obtained from Gazebo through the `/model/rod_payload/pose` topic published as a `geometry_msgs/PoseStamped` message. The controller extracts the translational coordinates directly from the message and converts the quaternion orientation into Euler angles to estimate the payload yaw and pitch states. The rod pose therefore acts as the central feedback signal for the closed-loop controller.

PID control structure Independent PID controllers were implemented for the payload states in the x, y, z, yaw, and pitch directions. The PID controller computes the correction based on the difference between the desired payload state and the measured payload state:

$$e(t) = r(t) - y(t)$$

where $r(t)$ is the desired state and $y(t)$ is the measured payload state.

The implemented PID law is given by:

$$u(t) = K_p e(t) + K_i \int e(t)\,dt + K_d \frac{de(t)}{dt}$$

where:

- K_p is the proportional gain,
- K_i is the integral gain,
- K_d is the derivative gain.

Adjusted payload target generation Instead of directly sending the desired payload pose to the UAVs, the PID controller first generates an adjusted payload target:

$$x_{adj} = x_{target} + u_x$$

$$y_{adj} = y_{target} + u_y$$

$$z_{adj} = z_{target} + u_z$$

Similarly, corrected yaw and pitch values are generated:

$$\psi_{adj} = \psi_{target} + u_\psi$$

$$\theta_{adj} = \theta_{target} + u_\theta$$

These adjusted states represent the corrected payload configuration after incorporating PID feedback compensation.

Geometric cooperative transformation The rod payload was modeled as a rigid body suspended between two UAVs. Once the corrected payload center and orientation are computed, the controller determines the required position of each UAV by calculating the rod endpoints geometrically.

Assuming the rod length is L, the displacement from the rod center to each UAV attachment point is:

$$dx = \frac{L}{2}\cos(\theta)\cos(\psi)$$

$$dy = \frac{L}{2}\cos(\theta)\sin(\psi)$$

$$dz = \frac{L}{2}\sin(\theta)$$

The desired positions for the two UAVs are then:

$$P_1 = \begin{bmatrix} x - dx \\ y - dy \\ z - dz \end{bmatrix}$$

$$P_2 = \begin{bmatrix} x + dx \\ y + dy \\ z + dz \end{bmatrix}$$

This transformation ensures that both UAVs cooperatively maintain the desired rod orientation and payload center simultaneously.

Coordinate frame conversion Gazebo internally operates in the ENU (East-North-Up) frame, whereas PX4 expects trajectory setpoints in the NED (North-East-Down) frame. Therefore, the computed payload endpoint positions were converted before transmission to PX4:

$$x_{NED} = y_{ENU}$$

$$y_{NED} = x_{ENU}$$

$$z_{NED} = -z_{ENU}$$

Additional positional offsets were added because the two drones were spawned at different initial positions inside the world environment.

PX4 offboard trajectory generation After the cooperative transformation, the controller publishes:

- `OffboardControlMode`
- `TrajectorySetpoint`
- `VehicleCommand`

messages independently for both PX4 instances. The setpoints are streamed continuously at approximately 50 Hz to ensure that PX4 remains in offboard mode without triggering failsafe timeout conditions. The controller first sends several initial setpoints before arming and enabling offboard mode, following PX4 offboard safety requirements.

Yaw synchronization To maintain coordinated motion during transport, both UAVs were commanded with a common yaw reference derived from the corrected payload yaw angle. An additional yaw offset of approximately 90° was introduced to align the PX4 body frame with the Gazebo payload orientation convention. This ensured proper heading alignment during forward transport motion.

Closed-loop cooperative behavior The overall control loop operates continuously at a timestep of approximately 20 ms. During each iteration:

1. Desired payload targets are received from the keyboard input node.
2. Actual payload pose is obtained from Gazebo.
3. PID corrections are computed.
4. Corrected payload pose is generated.
5. Geometric transformation computes UAV endpoint positions.
6. ENU-to-NED conversion is applied.
7. Cooperative trajectory setpoints are published to both PX4 controllers.

This architecture allows both drones to cooperatively stabilize and transport the rod payload while compensating for payload oscillation and tracking errors in real time.

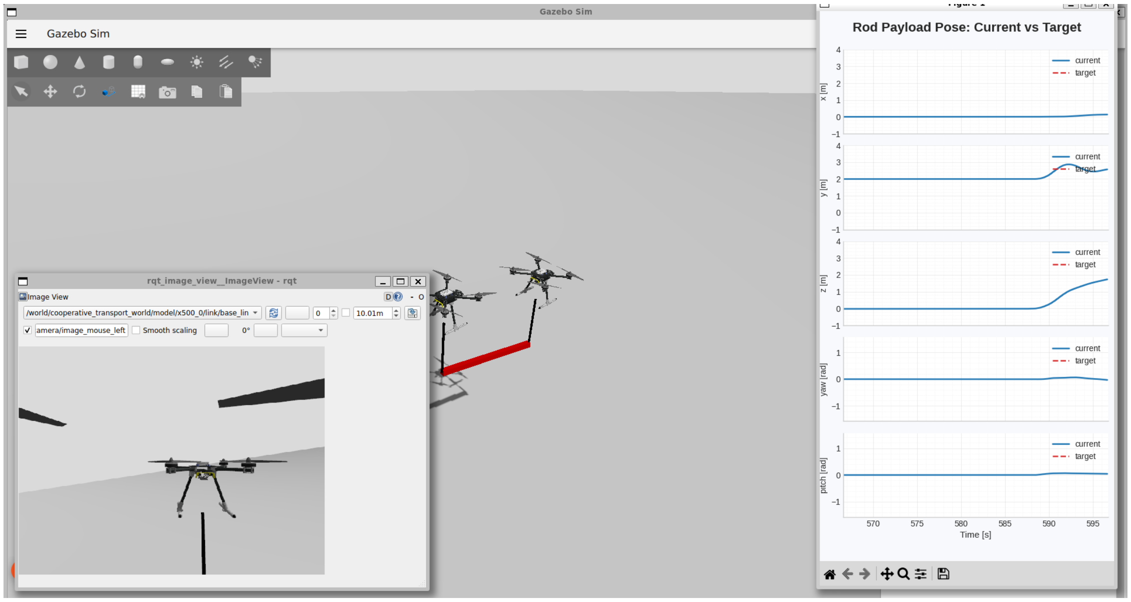

Figure 5: Gazebo Harmonic simulation with Camera-Based Rod Pose Estimation and Offboard Control Architecture.

Advantages of the implemented controller The implemented PID-based framework provides a computationally lightweight and modular baseline for cooperative payload transportation. Since the controller operates directly on payload pose feedback rather than independent drone trajectories, it naturally incorporates coupled payload dynamics into the control loop. Furthermore, the modular ROS 2 implementation enables future integration of:

- advanced nonlinear controllers,
- reinforcement learning policies,
- distributed cooperative control,
- adaptive gain tuning,
- vision-based state estimation, and
- fault-tolerant multi-UAV transport strategies.

4 Future prospects

The future scope of this project lies in extending the currently developed PID-based cooperative transportation framework toward an intelligent, robust, and learning-enabled multi-UAV aerial manipulation system. The present simulation and control architecture provides a modular foundation for integrating advanced cooperative control, state estimation, reinforcement learning, and communication-aware coordination strategies for transporting rod-shaped suspended payloads using multiple UAVs.

The next stage of the work will focus on the development of a **hierarchical cooperative control architecture** in which a centralized high-level controller generates intermediate objectives such as desired payload accelerations, cable tension distribution, and cooperative force allocation, while decentralized onboard PX4 controllers execute low-level attitude and position stabilization. This hierarchical separation is expected to improve scalability, modularity, and robustness of the overall system while maintaining compatibility with standard autopilot frameworks such as `PX4` and ROS 2.

A major future direction involves the integration of **safe reinforcement learning (Safe RL)** into the cooperative transportation pipeline. Instead of relying solely on fixed PID gains or model-dependent controllers, reinforcement learning policies will be explored for adaptive payload stabilization, swing suppression, cooperative force regulation, and trajectory tracking under uncertain operating conditions. Constrained reinforcement learning methods and safety-aware reward formulations will be incorporated to ensure bounded payload swing, stable cable tension, collision avoidance, and safe UAV operation during learning and deployment.

Another important extension of this work is the development of **learning-based nonlinear payload state estimation**. Accurate estimation of payload swing angles, angular velocity, rod orientation, and relative UAV-payload configuration is critical for stable cooperative transport. Future work will investigate multi-sensor fusion techniques using onboard IMUs, force sensors, camera feeds, and UAV odometry integrated through nonlinear observers and learning-enhanced estimation schemes. Both centralized and distributed state estimation approaches will be explored to reduce dependence on external positioning systems and improve robustness under noisy or partially observable environments.

The proposed framework will also be expanded to study **communication-aware cooperative control**. Different controller placement strategies, including onboard distributed control, centralized ground-station control, and hybrid architectures, will be evaluated. The effects of communication latency, packet loss, bandwidth limitations, and partial state sharing on cooperative payload transport performance will be analyzed in detail. This study is particularly important for real-world deployment scenarios where reliable inter-UAV communication cannot always be guaranteed.

Future work will additionally focus on improving the **payload dynamics modeling** of cable-suspended rod transportation systems. More accurate representations of cable flexibility, tension dynamics, aerodynamic disturbances, and inter-UAV coupling effects will be incorporated into the simulation and control framework. Environmental disturbances such as wind gusts, actuator uncertainty, and sensor noise will also be included to evaluate controller robustness under realistic operating conditions.

The currently developed ROS 2, Gazebo Harmonic, and PX4 SITL framework provides a suitable platform for validating these advanced algorithms in simulation before transitioning toward hardware implementation. Future validation will include extensive robustness testing under disturbances, communication delays, sensor noise, and dynamic trajectory tracking tasks using performance metrics such as payload swing amplitude, tracking error, cooperative force distribution, and energy efficiency.

In the longer term, the framework can be extended toward **real-world multi-UAV cooperative aerial manipulation applications** such as disaster relief logistics, infrastructure assembly, transportation of elongated industrial components, construction automation, and autonomous material handling in hazardous environments. The modular simulation and control architecture developed in this project can also serve as a foundation for future research in cooperative robotics, distributed autonomous systems, and intelligent aerial manipulation.

References

[1] J. Estevez, G. Garate, J. M. Lopez-Guede, and M. Larrea, "Review of aerial transportation of suspended-cable payloads with quadrotors," *Drones*, vol. 8, no. 2, p. 35, 2024.

[2] H. Li, H. Zhong, J. Gao, Y. Lv, J. Sha, J. Liang, H. Zhang, and Y. Wang, "A nonlinear trajectory tracking control strategy for quadrotor with suspended payload based on force sensor," *IEEE Transactions on Intelligent Vehicles*, vol. 9, no. 1, pp. 704–714, 2024.

[3] X. Wang and Y. Chen, "Cooperative Control for Rod-Shaped Payload Transportation with Two Quadrotors," in *Proceedings of the 2nd International Conference on Mechanical System Dynamics*, 2024.

[4] A. Asignacion, Jr. and S. Suzuki, "Historical and Current Landscapes of Autonomous Quadrotor Control: An Early-Career Researchers' Guide," *Drones*, vol. 8, no. 2, p. 44, 2024.

[5] D. Villa *et al.*, "Rod-shaped Payload Transportation Using Multiple Quadrotors," in *International Conference on Unmanned Aircraft Systems (ICUAS)*, 2019.

[6] Y. Xu *et al.*, "A Control Strategy for Swing Suppression in Dual-UAV Cooperative Rod Payload Transportation," in *5th International Conference on Control Theory and Applications (ICoCTA)*, 2025.

[7] P. Prajapati, A. Patidar, and V. Vashista, "A study on reinforcement learning based control of quadcopter with a cable-suspended payload," in *Proceedings of the 2023 6th International Conference on Advances in Robotics*, Ropar, India, jul 2023.

[8] Y. Ohira, A. Asignacion, and S. Suzuki, "Payload state estimation for cooperative manipulation using multiple uavs," in *16th International Micro Air Vehicle Conference and Competition*, San Andres Cholula, Puebla, Mexico, nov 2025, pp. 125–132, paper no. IMAV2025-16.

[9] E. L. de Angelis, "Swing angle estimation for multicopter slung load applications," *Aerospace Science and Technology*, vol. 89, pp. 264–274, 2019.

[10] Microsoft, "Project AirSim," 2023. [Online]. Available: https://www.microsoft.com/en-us/ai/autonomous-systems-project-airsim

[11] PX4 Development Team, "PX4 Autopilot," 2023. [Online]. Available: https://px4.io

[12] Open Robotics, "Robot Operating System (ROS)," 2023. [Online]. Available: https://www.ros.org

[13] ——, "Gazebo Simulator," 2023. [Online]. Available: https://gazebosim.org